Integrative analysis for sc-/sn-RNA-Seq Analysis in 10X Genomics data

Antonia Chroni for SJCRH DNB_BINF_Core

Contents


```
## The following objects are masked _by_ .GlobalEnv:
##
##     PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,
##     genome_name, integration_method, nfeatures_value, root_dir
```

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DNB Bioinformatics Core Pipeline: Standard sc-/sn-RNA-Seq Analysis in 10X Genomics data

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1 Information about this notebook

Biological heterogeneity in single-cell RNA-seq data is often confounded by technical factors including sequencing depth. The number of molecules detected in each cell can vary significantly between cells, even within the same celltype. Interpretation of scRNA-seq data requires effective pre-processing and normalization to remove this technical variability (for more information, see also scRNA-seq Dataset Integration[2]).

Hence, integration of single-cell sequencing datasets, for example across experimental batches, donors, or conditions, is often an important step in scRNA-seq workflows. Integrative analysis can help to match shared cell types and states across datasets, which can boost statistical power, and most importantly, facilitate accurate comparative analysis across datasets. Here, we include the following integration methods for integrative analysis:

(1) Seurat 'anchor-based'[3] integration workflow by Choudhary & Satija, 2022[4],
(2) Harmony[5] integration workflow by Korsunsky et al., 2019[6], and
(3) Liger[7] integration workflow by Welch et al., 2019[8].

Different integration algorithms, such as Seurat, Harmony, and Liger, often use different numbers of dimensions. The optimal number of dimensions depends on the specific algorithm and its associated parameters. While these algorithms generally consider similar variables, the mathematical approaches they use to process and transform those variables differ significantly, which means the same number of dimensions (or "resolutions") doesn't work equally well across all algorithms.

Even though they all refer to "resolution," the effect of resolution can vary from one algorithm to another, and the maximum number of dimensions they can effectively use may differ. Based on our experience, we recommend using 30 dimensions for Seurat, 30 for Harmony, and 20 for Liger. These settings tend to produce the most consistent integration results without unnecessarily increasing computational load.

2 Set up

```
attach(params)
```

```
## The following objects are masked _by_ .GlobalEnv:
##
##     PCA_Feature_List_value, Regress_Cell_Cycle_value, assay,
##     genome_name, integration_method, nfeatures_value, root_dir
```

```
suppressPackageStartupMessages({
  library(future)
  library(tidyverse)
  library(patchwork)
  library(Seurat)
  library(SeuratObject)
  library(harmony)
```

[2] https://www.singlecellcourse.org/scrna-seq-dataset-integration.html#seurat-v3-3-vs-5-10k-pbmc
[3] https://satijalab.org/seurat/articles/seurat5_integration
[4] https://doi.org/10.1186/s13059-021-02584-9
[5] https://portals.broadinstitute.org/harmony/
[6] https://www.nature.com/articles/s41592-019-0619-0
[7] https://welch-lab.github.io/liger/articles/liger_with_seurat.html
[8] https://doi.org/10.1016/j.cell.2019.05.006

```r
library(rliger)
library(RcppPlanc)
library(SeuratWrappers)
library(scooter)
library(reshape2)
library(RColorBrewer)
library(knitr)

# Evaluate Seurat R expressions asynchronously when possible using future package
options(future.globals.maxSize = future_globals_value)
plan(multisession, workers = parallelly::availableCores())})
```

3 Directories and paths to file Inputs/Outputs

```r
#analysis_dir <- file.path(root_dir, "analyses", "integrative-analysis")
#data_dir <- file.path(root_dir, "analyses", "upstream-analysis", "results", "04_Filter_obj
#upstream_analysis_dir <- file.path(root_dir, "analyses", "upstream-analysis")

# Input files
#data_file <- file.path(data_dir, "seurat_obj_merged_filtered.rds")

# Create results_dir
#results_dir <- file.path(analysis_dir, "results")
#if (!dir.exists(results_dir)) {
#  dir.create(results_dir)}

# Create plots directory
#report_dir <- file.path(analysis_dir, "plots")
#if (!dir.exists(report_dir)) {
#  dir.create(report_dir)}

#plots_dir <- file.path(report_dir, glue::glue("plots_{integration_method}"))
#if (!dir.exists(plots_dir)) {
#  dir.create(plots_dir)}

source(paste0(root_dir, "/figures/scripts/theme_plot.R"))
#source(paste0(analysis_dir, "/util/function-samples-integrate.R"))
#source(paste0(analysis_dir, "/util/custom-seurat-functions.R"))
#source(paste0(upstream_analysis_dir, "/util/function-process-Seurat.R"))

source(paste0(module_dir, "/util/function-samples-integrate.R"))
source(paste0(module_dir, "/util/custom-seurat-functions.R"))
source(paste0(module_dir, "/util/function-process-Seurat.R"))
```

4 Read seurat object

First, we will use the `seurat_obj_merged_filtered.rds` object as generated from the pipeline in the `upstream-analysis` module. This object contains all samples of the project merged.

```r
merged_obj <- readRDS(data_file)
```

5 Seurat 'anchor-based' integration workflow

Seurat integration creates a unified object that contains both original data ('RNA' assay) as well as integrated data ('integrated' assay).

5.1 Normal-sized datasets

SCT method (or sctransform) is a modeling framework for the normalization and variance stabilization of molecular count data from scRNA-seq experiments. This procedure omits the need for heuristic steps including pseudocount addition or log-transformation and improves common downstream analytical tasks such as variable gene selection, dimensional reduction, and differential expression. More specifically, the sctransform approach utilizes the Pearson residuals from negative binomial regression as input to standard dimensional reduction techniques.

5.2 Large-sized datasets

For very large datasets, the standard integration workflow can sometimes be prohibitively computationally expensive. In addition, Seurat and a lot of integration algorithms are designed to integrate data up to a maximum amount. Here, we use the `big_data` parameter as an option for integration of extremely large datasets. For example, 1.3 millions cells are too much and so this option exist for those datasets. The problem comes down to infinite matrix sizes and memory issues, no computer to data can compute this many variables so new algorithms exist to get around that through various methods. For more information, see Seurat v4 large data integration[9] and Seurat v5 large data integration[10].

A reference_list[11] can be used when conducting Seurat integration which help with speeding up integration.

Moreover, we improve efficiency and runtimes by employing (1) Reciprocal PCA (RPCA) and (2) Reference-based integration. The main efficiency improvements are gained in FindIntegrationAnchors(). First, we use reciprocal PCA (RPCA), to identify an effective space in which to find anchors. When determining anchors between any two datasets using reciprocal PCA, we project each dataset into the others PCA space and constrain the anchors by the same mutual neighborhood requirement. All downstream integration steps remain the same and we are able to 'correct' (or harmonize) the datasets.

Additionally, we use reference-based integration. In the standard workflow, we identify anchors between all pairs of datasets. While this gives datasets equal weight in downstream integration, it can also become computationally intensive. For example when integrating 10 different datasets, we perform 45 different pairwise comparisons. As an alternative, we introduce here the possibility of specifying one or more of the datasets as the 'reference' for integrated analysis, with the remainder designated as 'query' datasets. In this workflow, we do not identify anchors between pairs of query datasets, reducing the number of comparisons. For example, when integrating 10 datasets with one specified as a reference, we perform only 9 comparisons. Reference-based integration can be applied to either log-normalized or SCTransform-normalized datasets.

Note that the anchor-based RPCA integration represents a faster and more conservative (less correction) method for integration.

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[9] https://satijalab.org/seurat/archive/v4.3/integration_large_datasets
[10] https://satijalab.org/seurat/articles/seurat5_sketch_analysis
[11] https://satijalab.org/seurat/reference/findintegrationanchors#examples

```r
# We will use the Seurat 'anchor-based' workflow, if defined in `params`.
if (use_seurat_integration == "YES"){
  print_message <- "we use the Seurat 'anchor-based' workflow"

  # Make seurat_list
  seurat_obj_list <- list()
  seurat_obj_list <- SplitObject(merged_obj, split.by = "orig.ident")

  # Run function
  seurat_obj <- seurat_integration(seurat_obj_list = seurat_obj_list,
                                   nfeatures_value = nfeatures_value,
                                   num_dim_seurat = num_dim_seurat,
                                   num_dim_seurat_integration = num_dim_seurat_integration,
                                   big_data = big_data_value,
                                   Genome = genome_name,
                                   Regress_Cell_Cycle = Regress_Cell_Cycle_value,
                                   reference_list = reference_list_value,
                                   PCA_Feature_List = PCA_Feature_List_value)
  } else {
    print_message <- "we will skip usage of the Seurat 'anchor-based' workflow"}
```

Here, we will skip usage of the Seurat 'anchor-based' workflow.

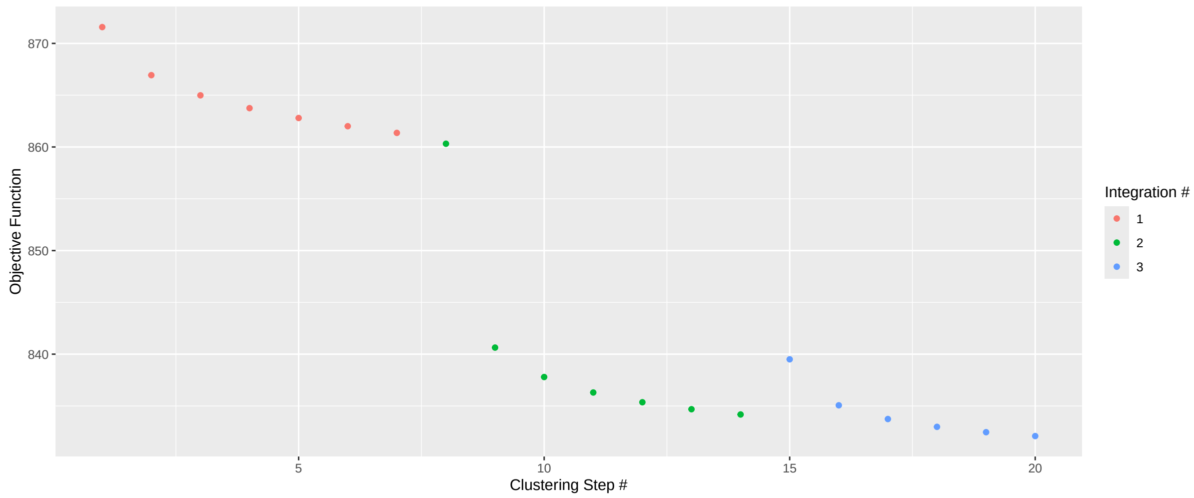
6 Harmony integration workflow

Harmony is a general-purpose R package with an efficient algorithm for integrating multiple data sets. It is especially useful for large single-cell datasets such as single-cell RNA-seq. Harmony algorithm projects cells into a shared embedding in which cells group by cell type rather than dataset-specific conditions. Harmony simultaneously accounts for multiple experimental and biological factors. We will run Harmony within our Seurat workflow.

```r
# We will use the Harmony integration workflow, if defined in `params`.
if (use_harmony_integration == "YES"){
  print_message <- "we use the Harmony integration workflow"
  seurat_obj <- harmony_integration(seurat_obj = merged_obj,
                                    variables_to_integrate = variable_value,
                                    num_dim = num_dim_harmony)
  # Plot
  name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
  p1 <- DimPlot(object = seurat_obj, reduction = glue::glue("{integration_method}"), pt.size
  p2 <- VlnPlot(object = seurat_obj, features = "harmony_1", group.by = "orig.ident", pt.si
  print(patchwork::wrap_plots(list(p1, p2), nrow = 1))
  ggsave(file = name, width = 12, height = 5, device = "png")

  } else {
    print_message <- "we will skip usage of the Harmony integration workflow"}
```

```
## Performing Harmony Integration

## Transposing data matrix

## Initializing state using k-means centroids initialization
```

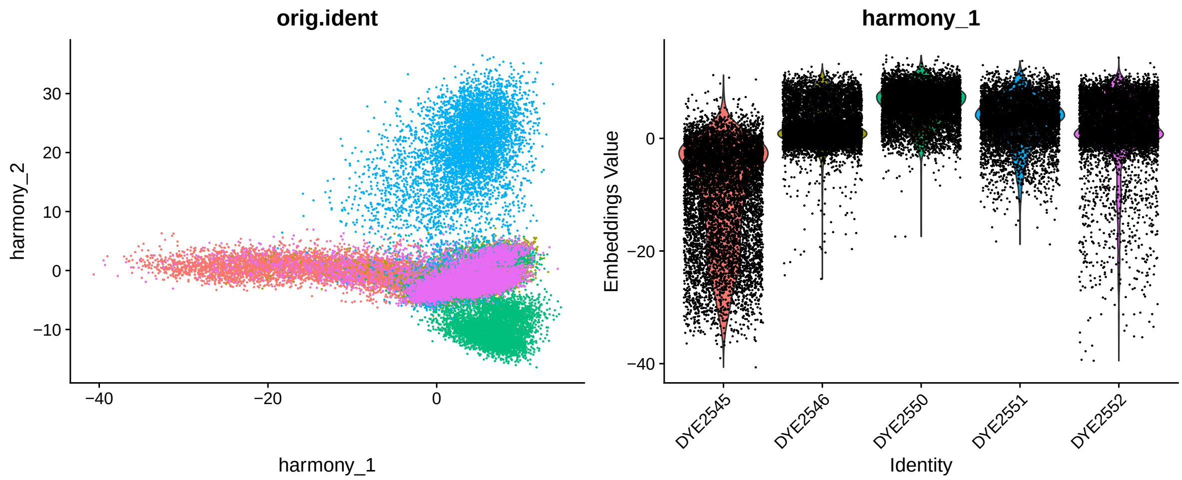
```
## Harmony 1/10

## Harmony 2/10

## Harmony 3/10

## Harmony converged after 3 iterations
```




```
## Performing Uniform Manifold Approximation and Projection (UMAP) dimensional reduction tec

## 18:45:40 UMAP embedding parameters a = 0.9922 b = 1.112

## 18:45:40 Read 35718 rows and found 30 numeric columns

## 18:45:40 Using Annoy for neighbor search, n_neighbors = 30

## 18:45:40 Building Annoy index with metric = cosine, n_trees = 50

## 0%   10   20   30   40   50   60   70   80   90   100%

## [----|----|----|----|----|----|----|----|----|----|

## **************************************************|
## 18:45:44 Writing NN index file to temp file /lsf_tmp/264625707.tmpdir/RtmpbhtMvE/file2f98
## 18:45:44 Searching Annoy index using 16 threads, search_k = 3000
## 18:45:45 Annoy recall = 100%
## 18:45:45 Commencing smooth kNN distance calibration using 16 threads with target n_neighb
## 18:45:46 Initializing from normalized Laplacian + noise (using irlba)
## 18:45:47 Commencing optimization for 200 epochs, with 1513606 positive edges
## 18:46:02 Optimization finished

## Generate and add metadata

##
##
##  ========== saving metadata ==========
##
##
##
##
##  ========== saving metadata ==========

## Computing the k.param nearest neighbors for a given dataset and clusters
```

```
## Computing nearest neighbor graph
## Computing SNN
```

```
## Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck
##
## Number of nodes: 35718
## Number of edges: 1200086
##
## Running Louvain algorithm...
## Maximum modularity in 10 random starts: 0.8811
## Number of communities: 17
## Elapsed time: 8 seconds
## Returned Seurat object
```



Here, we use the Harmony integration workflow.

7 Liger integration workflow

LIGER is an algorithm that enables the identification of shared cell types across individuals, species, and multiple modalities (gene expression, epigenetic or spatial data), as well as dataset-specific features, offering a unified analysis of heterogeneous single-cell datasets.

More specifically, LIGER can be used to integrate datasets: (1) from multiple experiments and biological contexts (differences in technical variation, e.g., number of cells sampled, or in the depth of sequencing allocated to each cell), (2) across space (in situ and single-cell expression data to spatially locate fine subtypes of cells) and (3) various modalities (single-cell RNA-seq and DNA methylation or epigenomes). Across modalities, datasets may vary widely in dynamic range (gene expression versus chromatin accessibility), direction of relationship (RNA-seq versus DNA methylation), or in the number of genes measured (targeted quantification versus unbiased approaches).

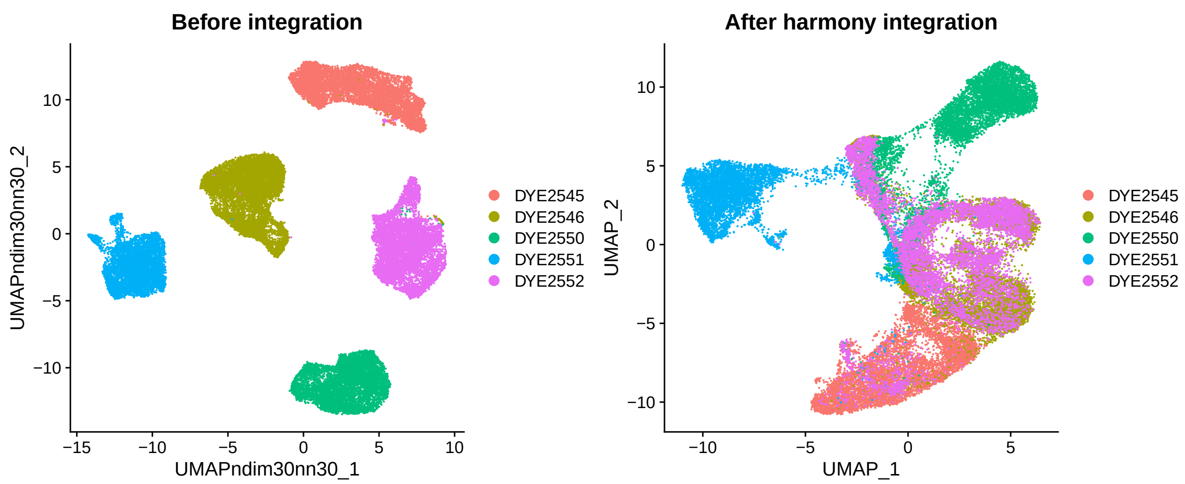
```r
# We will use the Liger workflow, if defined in `params`.
if (use_liger_integration == "YES"){
  print_message <- "we use the Liger workflow"
  seurat_obj <- liger_integration(seurat_obj = merged_obj,
                                  variables_to_integrate = variable_value,
                                  n_neighbors = n_neighbors_value)
```

```
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_{integration_method}.png"))
DimPlot(seurat_obj, reduction = glue::glue("{integration_method}"), group.by = c("orig.ide
ggsave(file = name, width = 6, height = 5, device = "png")

} else {
  print_message <- "we will skip usage of the Liger workflow"}
```

Here, we will skip usage of the Liger workflow.

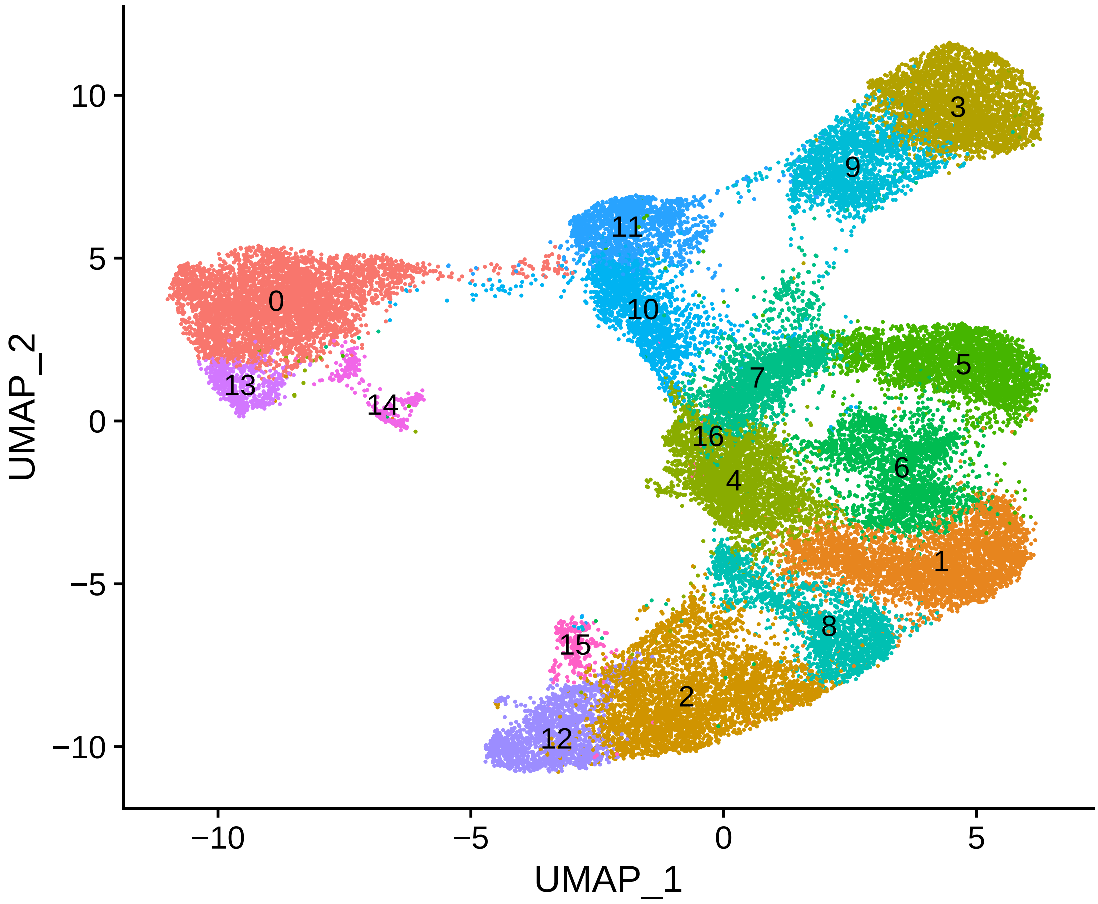
8 Before vs After integration plot

```
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_unitegrated_vs_{integration_method}.png"))
p1 <- DimPlot(merged_obj, group.by = c("orig.ident")) + ggtitle("Before integration")
p2 <- DimPlot(seurat_obj, reduction = "umap", group.by = c("orig.ident")) + ggtitle(glue::gl
patchwork::wrap_plots(list(p1, p2), nrow = 1)
```




```
ggsave(file = name, width = 12, height = 5, device = "png")
```

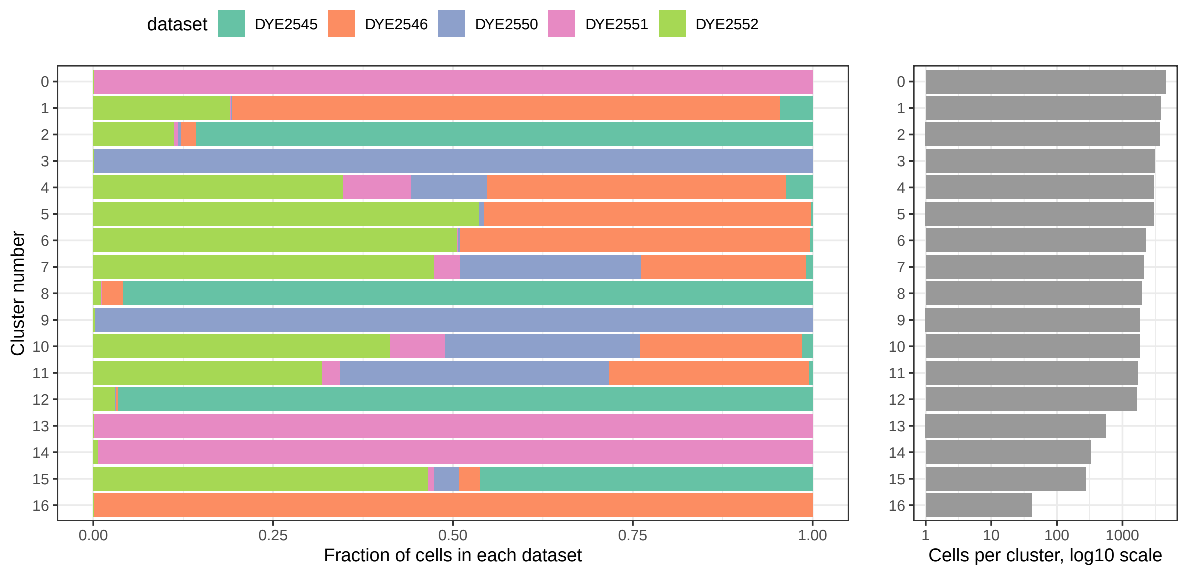
```
seurat_obj <- SetIdent(seurat_obj, value = "seurat_clusters")

# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_clusters_{integration_method}.png"))
DimPlot(seurat_obj, reduction = "umap", label = TRUE) + NoLegend()
```




```r
ggsave(file = name, width = 6, height = 5, device = "png")
```

```r
# Plot
name <- paste0(plots_dir, "/", glue::glue("plot_fraction_of_cells_{integration_method}.png"
plot_integrated_clusters(seurat_obj)
```

```
## Using cluster as id variables
```



```r
ggsave(file = name, width = 10, height = 6, device = "png")
```

9 Save output files

```r
# Identify columns with a '.1' suffix
cols_to_remove <- grep("\\.1$", colnames(seurat_obj@meta.data), value = TRUE)

# Exclude columns that match the specific patterns (e.g., {assay}_snn_res.0.1, {assay}_snn_
cols_to_remove <- cols_to_remove[!grepl(glue::glue("^{assay}_snn_res\\.0\\.1$"), cols_to_rem
                                 !grepl(glue::glue("^{assay}_snn_res\\.1$"), cols_to_rem
                                 !grepl(glue::glue("^{assay}_snn_res\\.10$"), cols_to_re

# Remove the columns
seurat_obj@meta.data <- seurat_obj@meta.data[, !colnames(seurat_obj@meta.data) %in% cols_to_
head(seurat_obj@meta.data)
```

```
##                             orig.ident nCount_RNA nFeature_RNA percent.mito
## DYE2545:AAACCCAAGATGTTCC    DYE2545      22226         5185        0.972
## DYE2545:AAACCCAAGGAAAGAC    DYE2545      34387         7304        1.658
## DYE2545:AAACCCAAGGCTCTCG    DYE2545       8775         3123        2.405
## DYE2545:AAACCCACAAACCACT    DYE2545       7725         3287        1.553
## DYE2545:AAACCCACAAGTGGTG    DYE2545      14377         4845        1.934
## DYE2545:AAACCCACAATACAGA    DYE2545       1609         1078        2.237
##                             log10GenesPerUMI      ID          SAMPLE
## DYE2545:AAACCCAAGATGTTCC           0.8545818 DYE2545 3251768_DYE2545
## DYE2545:AAACCCAAGGAAAGAC           0.8516810 DYE2545 3251768_DYE2545
## DYE2545:AAACCCAAGGCTCTCG           0.8862168 DYE2545 3251768_DYE2545
## DYE2545:AAACCCACAAACCACT           0.9045503 DYE2545 3251768_DYE2545
## DYE2545:AAACCCACAAGTGGTG           0.8863848 DYE2545 3251768_DYE2545
## DYE2545:AAACCCACAATACAGA           0.9457557 DYE2545 3251768_DYE2545
##
## DYE2545:AAACCCAAGATGTTCC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCAAGGAAAGAC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCAAGGCTCTCG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCACAAACCACT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCACAAGTGGTG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCACAATACAGA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
##                                   SJUID        SJID   cancer_type
## DYE2545:AAACCCAAGATGTTCC SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCAAGGAAAGAC SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCAAGGCTCTCG SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCACAAACCACT SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCACAAGTGGTG SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCACAATACAGA SJHBBBFP3VC SJNBL012407 Neuroblastoma
```

```
##                            Subject_Disease_Code Subject_Disease_Subtype_Code
## DYE2545:AAACCCAAGATGTTCC                    NBL                          NBL
## DYE2545:AAACCCAAGGAAAGAC                    NBL                          NBL
## DYE2545:AAACCCAAGGCTCTCG                    NBL                          NBL
## DYE2545:AAACCCACAAACCACT                    NBL                          NBL
## DYE2545:AAACCCACAAGTGGTG                    NBL                          NBL
## DYE2545:AAACCCACAATACAGA                    NBL                          NBL
##                            tissue_type assay_technology_updated assay_technology
## DYE2545:AAACCCAAGATGTTCC    CCLF_cells              10x_scRNAseq     10x scRNAseq
## DYE2545:AAACCCAAGGAAAGAC    CCLF_cells              10x_scRNAseq     10x scRNAseq
## DYE2545:AAACCCAAGGCTCTCG    CCLF_cells              10x_scRNAseq     10x scRNAseq
## DYE2545:AAACCCACAAACCACT    CCLF_cells              10x_scRNAseq     10x scRNAseq
## DYE2545:AAACCCACAAGTGGTG    CCLF_cells              10x_scRNAseq     10x scRNAseq
## DYE2545:AAACCCACAATACAGA    CCLF_cells              10x_scRNAseq     10x scRNAseq
##                            historic_data RPMI genome_reference_updated_code
## DYE2545:AAACCCAAGATGTTCC              no   no                  human-genome
## DYE2545:AAACCCAAGGAAAGAC              no   no                  human-genome
## DYE2545:AAACCCAAGGCTCTCG              no   no                  human-genome
## DYE2545:AAACCCACAAACCACT              no   no                  human-genome
## DYE2545:AAACCCACAAGTGGTG              no   no                  human-genome
## DYE2545:AAACCCACAATACAGA              no   no                  human-genome
##                            genome_reference_updated genome_reference
## DYE2545:AAACCCAAGATGTTCC                dual-genome      dual-genome
## DYE2545:AAACCCAAGGAAAGAC                dual-genome      dual-genome
## DYE2545:AAACCCAAGGCTCTCG                dual-genome      dual-genome
## DYE2545:AAACCCACAAACCACT                dual-genome      dual-genome
## DYE2545:AAACCCACAAGTGGTG                dual-genome      dual-genome
## DYE2545:AAACCCACAATACAGA                dual-genome      dual-genome
##
## DYE2545:AAACCCAAGATGTTCC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCAAGGAAAGAC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCAAGGCTCTCG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCACAAACCACT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCACAAGTGGTG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCACAATACAGA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
##                          DYExxxx DYE_Melody_unsure DYExxxx_corrected
## DYE2545:AAACCCAAGATGTTCC DYE2545              <NA>           DYE2545
## DYE2545:AAACCCAAGGAAAGAC DYE2545              <NA>           DYE2545
## DYE2545:AAACCCAAGGCTCTCG DYE2545              <NA>           DYE2545
## DYE2545:AAACCCACAAACCACT DYE2545              <NA>           DYE2545
## DYE2545:AAACCCACAAGTGGTG DYE2545              <NA>           DYE2545
## DYE2545:AAACCCACAATACAGA DYE2545              <NA>           DYE2545
##                          SRM_updated    SRM SRM_Sample_ID_updated SRM_Sample_ID
## DYE2545:AAACCCAAGATGTTCC      841877 841877               3251768       3251768
## DYE2545:AAACCCAAGGAAAGAC      841877 841877               3251768       3251768
```

```
## DYE2545:AAACCCAAGGCTCTCG      841877 841877             3251768       3251768
## DYE2545:AAACCCACAAACCACT      841877 841877             3251768       3251768
## DYE2545:AAACCCACAAGTGGTG      841877 841877             3251768       3251768
## DYE2545:AAACCCACAATACAGA      841877 841877             3251768       3251768
##                             Experimental_Sample sample_3D
## DYE2545:AAACCCAAGATGTTCC SJ1041CCLF 01.24.003        No
## DYE2545:AAACCCAAGGAAAGAC SJ1041CCLF 01.24.003        No
## DYE2545:AAACCCAAGGCTCTCG SJ1041CCLF 01.24.003        No
## DYE2545:AAACCCACAAACCACT SJ1041CCLF 01.24.003        No
## DYE2545:AAACCCACAAGTGGTG SJ1041CCLF 01.24.003        No
## DYE2545:AAACCCACAATACAGA SJ1041CCLF 01.24.003        No
##                          sample_2D.NOT.NEEDED.2025.04.10 Tissue_Bank_ID
## DYE2545:AAACCCAAGATGTTCC                              No     TB-13-3627
## DYE2545:AAACCCAAGGAAAGAC                              No     TB-13-3627
## DYE2545:AAACCCAAGGCTCTCG                              No     TB-13-3627
## DYE2545:AAACCCACAAACCACT                              No     TB-13-3627
## DYE2545:AAACCCACAAGTGGTG                              No     TB-13-3627
## DYE2545:AAACCCACAATACAGA                              No     TB-13-3627
##                          Xenograft_CompBio_ID CSTN_MAST_ID                 Date
## DYE2545:AAACCCAAGATGTTCC       SJNBL012407_X1       MAST97 2024-12-16T00:00:00Z
## DYE2545:AAACCCAAGGAAAGAC       SJNBL012407_X1       MAST97 2024-12-16T00:00:00Z
## DYE2545:AAACCCAAGGCTCTCG       SJNBL012407_X1       MAST97 2024-12-16T00:00:00Z
## DYE2545:AAACCCACAAACCACT       SJNBL012407_X1       MAST97 2024-12-16T00:00:00Z
## DYE2545:AAACCCACAAGTGGTG       SJNBL012407_X1       MAST97 2024-12-16T00:00:00Z
## DYE2545:AAACCCACAATACAGA       SJNBL012407_X1       MAST97 2024-12-16T00:00:00Z
##                          SRM_Submission_Information
## DYE2545:AAACCCAAGATGTTCC       SJ1041CCLF 01.24.003
## DYE2545:AAACCCAAGGAAAGAC       SJ1041CCLF 01.24.003
## DYE2545:AAACCCAAGGCTCTCG       SJ1041CCLF 01.24.003
## DYE2545:AAACCCACAAACCACT       SJ1041CCLF 01.24.003
## DYE2545:AAACCCACAAGTGGTG       SJ1041CCLF 01.24.003
## DYE2545:AAACCCACAATACAGA       SJ1041CCLF 01.24.003
##                          Compbioname_for_publications     Comp_Bio_Sample_Name
## DYE2545:AAACCCAAGATGTTCC               SJNBL012407_C3 SJNBL012407_C3-
SJ1041T_2
## DYE2545:AAACCCAAGGAAAGAC               SJNBL012407_C3 SJNBL012407_C3-
SJ1041T_2
## DYE2545:AAACCCAAGGCTCTCG               SJNBL012407_C3 SJNBL012407_C3-
SJ1041T_2
## DYE2545:AAACCCACAAACCACT               SJNBL012407_C3 SJNBL012407_C3-
SJ1041T_2
## DYE2545:AAACCCACAAGTGGTG               SJNBL012407_C3 SJNBL012407_C3-
SJ1041T_2
## DYE2545:AAACCCACAATACAGA               SJNBL012407_C3 SJNBL012407_C3-
SJ1041T_2
##                          Comp_Bio_Subject_ID Preferred_Sample_ID
## DYE2545:AAACCCAAGATGTTCC            SJ012407           SJ1041T_2
## DYE2545:AAACCCAAGGAAAGAC            SJ012407           SJ1041T_2
## DYE2545:AAACCCAAGGCTCTCG            SJ012407           SJ1041T_2
## DYE2545:AAACCCACAAACCACT            SJ012407           SJ1041T_2
## DYE2545:AAACCCACAAGTGGTG            SJ012407           SJ1041T_2
```

```
## DYE2545:AAACCCACAATACAGA            SJ012407          SJ1041T_2
##                            Sample_Type_Code Prior_Comp_Bio_ID PATH NOTES
## DYE2545:AAACCCAAGATGTTCC                C       SJNBL012407 <NA>  <NA>
## DYE2545:AAACCCAAGGAAAGAC                C       SJNBL012407 <NA>  <NA>
## DYE2545:AAACCCAAGGCTCTCG                C       SJNBL012407 <NA>  <NA>
## DYE2545:AAACCCACAAACCACT                C       SJNBL012407 <NA>  <NA>
## DYE2545:AAACCCACAAGTGGTG                C       SJNBL012407 <NA>  <NA>
## DYE2545:AAACCCACAATACAGA                C       SJNBL012407 <NA>  <NA>
##                            match_SJID_RPMI     S.Score  G2M.Score Phase    UMAP_1
## DYE2545:AAACCCAAGATGTTCC  SJNBL012407_no -0.22486652 -0.2675636    G1  9.085792
## DYE2545:AAACCCAAGGAAAGAC  SJNBL012407_no -0.05736943 -0.1675573    G1  6.835735
## DYE2545:AAACCCAAGGCTCTCG  SJNBL012407_no -0.20725037 -0.1973390    G1  5.306400
## DYE2545:AAACCCACAAACCACT  SJNBL012407_no -0.28974713 -0.2804960    G1  4.128226
## DYE2545:AAACCCACAAGTGGTG  SJNBL012407_no -0.31975873 -0.3241192    G1  4.214514
## DYE2545:AAACCCACAATACAGA  SJNBL012407_no -0.11510892 -0.2054158    G1 -
3.577932
##                               UMAP_2       PC_1        PC_2      PC_3       PC_4
## DYE2545:AAACCCAAGATGTTCC -1.5443104 -15.863874 -32.6546839 0.8260766 -
18.296167
## DYE2545:AAACCCAAGGAAAGAC -2.2798051  -9.431601 -21.4253276 1.4509622 -
15.964464
## DYE2545:AAACCCAAGGCTCTCG -2.2335602  -9.189331 -11.7659123 2.8201872 -
12.901577
## DYE2545:AAACCCACAAACCACT -0.2242093  -0.632111  -8.1764752 4.6344284 -
11.745347
## DYE2545:AAACCCACAAGTGGTG -1.3846704  -3.746334  -7.1768090 2.1580190 -
12.186304
## DYE2545:AAACCCACAATACAGA  2.3944530  -4.619726   0.4027877 3.6466166  -
5.877808
##                                PC_5      PC_6       PC_7      PC_8       PC_9
## DYE2545:AAACCCAAGATGTTCC 7.4672071 -6.495609  6.9242178 10.356101 -3.1697889
## DYE2545:AAACCCAAGGAAAGAC 2.2837768 -5.869902  3.2564196  3.783053 -4.3649521
## DYE2545:AAACCCAAGGCTCTCG 3.0042420  1.099606 -1.0354393 -4.115567 -2.4106450
## DYE2545:AAACCCACAAACCACT 1.4077577  1.638154 -0.9564433 -7.009008 -3.0709138
## DYE2545:AAACCCACAAGTGGTG 0.3765286  1.134238 -0.7682338 -7.173833 -2.4168836
## DYE2545:AAACCCACAATACAGA 3.7909257  5.151632 -2.8199820 -5.585466  0.1286505
##                                 PC_10     PC_11     PC_12      PC_13      PC_14
## DYE2545:AAACCCAAGATGTTCC -22.8791552  2.318805  9.863862 -7.4874913 10.7100504
## DYE2545:AAACCCAAGGAAAGAC   0.4776367  1.408204 -2.874774  3.5337765 -
2.1880440
## DYE2545:AAACCCAAGGCTCTCG   6.4005139 -2.511625 -4.417163  3.4626152 -
5.3307465
## DYE2545:AAACCCACAAACCACT  13.9329421 -3.885880  2.272538 -0.6860176  3.0788362
## DYE2545:AAACCCACAAGTGGTG  10.5425499 -1.590351 -2.827610 -1.4508638  2.0134072
## DYE2545:AAACCCACAATACAGA   5.3019733 -3.024046  2.045094 -1.1220299  0.4380948
##                                PC_15      PC_16      PC_17       PC_18
## DYE2545:AAACCCAAGATGTTCC -6.3503098  2.2426751  1.8916413  0.75978639
## DYE2545:AAACCCAAGGAAAGAC  1.2392725 -1.2604325 -0.9685792 -3.78744852
## DYE2545:AAACCCAAGGCTCTCG  3.8559767 -1.1604766 -1.7994313 -2.49335533
## DYE2545:AAACCCACAAACCACT -1.7560677  0.6154134  1.6720582  0.07299006
## DYE2545:AAACCCACAAGTGGTG -0.3433200 -1.1592913  2.1567640 -4.65308816
```

```
## DYE2545:AAACCCACAATACAGA -0.5391842  0.2028160 -0.4331900  0.95269650
##                                 PC_19      PC_20       PC_21      PC_22
## DYE2545:AAACCCAAGATGTTCC -0.788243693 -2.2202915 -2.38143583 -2.9730362
## DYE2545:AAACCCAAGGAAAGAC  0.692745572  3.5548754  2.98895923  2.4440021
## DYE2545:AAACCCAAGGCTCTCG -1.601766611 -0.9419118  2.85820635  0.3180130
## DYE2545:AAACCCACAAACCACT  1.929624380  0.7687017 -0.08372203  0.4231828
## DYE2545:AAACCCACAAGTGGTG -2.328754025 -1.1156528 -1.02724389  2.5277115
## DYE2545:AAACCCACAATACAGA  0.005967839 -1.6297936 -0.53810839 -1.0408624
##                               PC_23      PC_24       PC_25      PC_26
## DYE2545:AAACCCAAGATGTTCC -0.9262151  3.5626954  6.52893164 -2.1101894
## DYE2545:AAACCCAAGGAAAGAC  0.9979305 -0.5610950  0.04656445  1.5751073
## DYE2545:AAACCCAAGGCTCTCG -0.5923591 -0.9212251 -3.33533561  0.4304649
## DYE2545:AAACCCACAAACCACT -0.7862789 -1.2788869 -1.49469591  0.2314583
## DYE2545:AAACCCACAAGTGGTG -1.7370136 -0.2650034 -2.29214870 -0.4044628
## DYE2545:AAACCCACAATACAGA -2.2783067  0.5782148 -1.53033807  0.1998572
##                               PC_27       PC_28      PC_29      PC_30
## DYE2545:AAACCCAAGATGTTCC  0.5164932  2.65092355 -6.4706949 -2.1001067
## DYE2545:AAACCCAAGGAAAGAC -2.6024575  0.36146655  0.6870124  0.8580130
## DYE2545:AAACCCAAGGCTCTCG -0.9915104  0.66481255 -3.3209920  4.7293540
## DYE2545:AAACCCACAAACCACT  0.8904170  0.01733426  1.4488693 -2.5102127
## DYE2545:AAACCCACAAGTGGTG -0.2296887 -1.35753530  0.8580106  2.0390308
## DYE2545:AAACCCACAATACAGA  0.8766829 -1.53763292  1.8993305  0.0598051
##                          RNA_snn_res.0 seurat_clusters RNA_snn_res.0.1
## DYE2545:AAACCCAAGATGTTCC             1              12               3
## DYE2545:AAACCCAAGGAAAGAC             1              15               3
## DYE2545:AAACCCAAGGCTCTCG             1               2               3
## DYE2545:AAACCCACAAACCACT             1               8               3
## DYE2545:AAACCCACAAGTGGTG             1               2               3
## DYE2545:AAACCCACAATACAGA             1               4               2
##                          RNA_snn_res.0.2 RNA_snn_res.0.3 RNA_snn_res.0.4
## DYE2545:AAACCCAAGATGTTCC               4               5               6
## DYE2545:AAACCCAAGGAAAGAC               4               5               6
## DYE2545:AAACCCAAGGCTCTCG               4               5               7
## DYE2545:AAACCCACAAACCACT               4               5               7
## DYE2545:AAACCCACAAGTGGTG               4               5               7
## DYE2545:AAACCCACAATACAGA               2               1               2
##                          RNA_snn_res.0.5 RNA_snn_res.0.6 RNA_snn_res.0.7
## DYE2545:AAACCCAAGATGTTCC               6               8               9
## DYE2545:AAACCCAAGGAAAGAC               6               8               9
## DYE2545:AAACCCAAGGCTCTCG               7               5               7
## DYE2545:AAACCCACAAACCACT               7               5               7
## DYE2545:AAACCCACAAGTGGTG               7               5               7
## DYE2545:AAACCCACAATACAGA               1               1               2
##                          RNA_snn_res.0.8 RNA_snn_res.0.9 RNA_snn_res.1
## DYE2545:AAACCCAAGATGTTCC              12               8             8
## DYE2545:AAACCCAAGGAAAGAC              15               8             8
## DYE2545:AAACCCAAGGCTCTCG               2               6             6
## DYE2545:AAACCCACAAACCACT               8               6             6
## DYE2545:AAACCCACAAGTGGTG               2               6             6
## DYE2545:AAACCCACAATACAGA               4               2             2
##                          RNA_snn_res.2 RNA_snn_res.3 RNA_snn_res.4
```

```
## DYE2545:AAACCCAAGATGTTCC            23            26            23
## DYE2545:AAACCCAAGGAAAGAC            11             6             6
## DYE2545:AAACCCAAGGCTCTCG             1            20            22
## DYE2545:AAACCCACAAACCACT             1             7            20
## DYE2545:AAACCCACAAGTGGTG             1             7            20
## DYE2545:AAACCCACAATACAGA             2             1             4
##                          RNA_snn_res.5 RNA_snn_res.6 RNA_snn_res.7
## DYE2545:AAACCCAAGATGTTCC            19             8             4
## DYE2545:AAACCCAAGGAAAGAC            10             5            15
## DYE2545:AAACCCAAGGCTCTCG            32            29            25
## DYE2545:AAACCCACAAACCACT            14            13            16
## DYE2545:AAACCCACAAGTGGTG            14            13            16
## DYE2545:AAACCCACAATACAGA             2             1             1
##                          RNA_snn_res.8 RNA_snn_res.9 RNA_snn_res.10
## DYE2545:AAACCCAAGATGTTCC            19            15             15
## DYE2545:AAACCCAAGGAAAGAC             6            11              7
## DYE2545:AAACCCAAGGCTCTCG            20            19             54
## DYE2545:AAACCCACAAACCACT            52            48             44
## DYE2545:AAACCCACAAGTGGTG            52            48             44
## DYE2545:AAACCCACAATACAGA            32             1             17
##                          orig.ident.1 nCount_RNA.1 nFeature_RNA.1
## DYE2545:AAACCCAAGATGTTCC      DYE2545        22226           5185
## DYE2545:AAACCCAAGGAAAGAC      DYE2545        34387           7304
## DYE2545:AAACCCAAGGCTCTCG      DYE2545         8775           3123
## DYE2545:AAACCCACAAACCACT      DYE2545         7725           3287
## DYE2545:AAACCCACAAGTGGTG      DYE2545        14377           4845
## DYE2545:AAACCCACAATACAGA      DYE2545         1609           1078
##                          percent.mito.1 log10GenesPerUMI.1    ID.1
## DYE2545:AAACCCAAGATGTTCC          0.972          0.8545818 DYE2545
## DYE2545:AAACCCAAGGAAAGAC          1.658          0.8516810 DYE2545
## DYE2545:AAACCCAAGGCTCTCG          2.405          0.8862168 DYE2545
## DYE2545:AAACCCACAAACCACT          1.553          0.9045503 DYE2545
## DYE2545:AAACCCACAAGTGGTG          1.934          0.8863848 DYE2545
## DYE2545:AAACCCACAATACAGA          2.237          0.9457557 DYE2545
##                                 SAMPLE.1
## DYE2545:AAACCCAAGATGTTCC 3251768_DYE2545
## DYE2545:AAACCCAAGGAAAGAC 3251768_DYE2545
## DYE2545:AAACCCAAGGCTCTCG 3251768_DYE2545
## DYE2545:AAACCCACAAACCACT 3251768_DYE2545
## DYE2545:AAACCCACAAGTGGTG 3251768_DYE2545
## DYE2545:AAACCCACAATACAGA 3251768_DYE2545
##
## DYE2545:AAACCCAAGATGTTCC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCAAGGAAAGAC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCAAGGCTCTCG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCACAAACCACT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
## DYE2545:AAACCCACAAGTGGTG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
```

```
1/3251768/
## DYE2545:AAACCCACAATACAGA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/3251768/
##                              SJUID.1      SJID.1 cancer_type.1
## DYE2545:AAACCCAAGATGTTCC SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCAAGGAAAGAC SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCAAGGCTCTCG SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCACAAACCACT SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCACAAGTGGTG SJHBBBFP3VC SJNBL012407 Neuroblastoma
## DYE2545:AAACCCACAATACAGA SJHBBBFP3VC SJNBL012407 Neuroblastoma
##                          Subject_Disease_Code.1 Subject_Disease_Subtype_Code.1
## DYE2545:AAACCCAAGATGTTCC                    NBL                            NBL
## DYE2545:AAACCCAAGGAAAGAC                    NBL                            NBL
## DYE2545:AAACCCAAGGCTCTCG                    NBL                            NBL
## DYE2545:AAACCCACAAACCACT                    NBL                            NBL
## DYE2545:AAACCCACAAGTGGTG                    NBL                            NBL
## DYE2545:AAACCCACAATACAGA                    NBL                            NBL
##                          tissue_type.1 assay_technology_updated.1
## DYE2545:AAACCCAAGATGTTCC    CCLF_cells               10x_scRNAseq
## DYE2545:AAACCCAAGGAAAGAC    CCLF_cells               10x_scRNAseq
## DYE2545:AAACCCAAGGCTCTCG    CCLF_cells               10x_scRNAseq
## DYE2545:AAACCCACAAACCACT    CCLF_cells               10x_scRNAseq
## DYE2545:AAACCCACAAGTGGTG    CCLF_cells               10x_scRNAseq
## DYE2545:AAACCCACAATACAGA    CCLF_cells               10x_scRNAseq
##                          assay_technology.1 historic_data.1 RPMI.1
## DYE2545:AAACCCAAGATGTTCC       10x scRNAseq              no     no
## DYE2545:AAACCCAAGGAAAGAC       10x scRNAseq              no     no
## DYE2545:AAACCCAAGGCTCTCG       10x scRNAseq              no     no
## DYE2545:AAACCCACAAACCACT       10x scRNAseq              no     no
## DYE2545:AAACCCACAAGTGGTG       10x scRNAseq              no     no
## DYE2545:AAACCCACAATACAGA       10x scRNAseq              no     no
##                          genome_reference_updated_code.1
## DYE2545:AAACCCAAGATGTTCC                    human-genome
## DYE2545:AAACCCAAGGAAAGAC                    human-genome
## DYE2545:AAACCCAAGGCTCTCG                    human-genome
## DYE2545:AAACCCACAAACCACT                    human-genome
## DYE2545:AAACCCACAAGTGGTG                    human-genome
## DYE2545:AAACCCACAATACAGA                    human-genome
##                          genome_reference_updated.1 genome_reference.1
## DYE2545:AAACCCAAGATGTTCC                dual-genome        dual-genome
## DYE2545:AAACCCAAGGAAAGAC                dual-genome        dual-genome
## DYE2545:AAACCCAAGGCTCTCG                dual-genome        dual-genome
## DYE2545:AAACCCACAAACCACT                dual-genome        dual-genome
## DYE2545:AAACCCACAAGTGGTG                dual-genome        dual-genome
## DYE2545:AAACCCACAATACAGA                dual-genome        dual-genome
##
## DYE2545:AAACCCAAGATGTTCC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCAAGGAAAGAC /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCAAGGCTCTCG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
```

```
1/
## DYE2545:AAACCCACAAACCACT /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCACAAGTGGTG /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
## DYE2545:AAACCCACAATACAGA /research/groups/dyergrp/projects/dyergrp_hartwell/common/illumi
1/
##                          DYExxxx.1 DYE_Melody_unsure.1 DYExxxx_corrected.1
## DYE2545:AAACCCAAGATGTTCC   DYE2545                <NA>             DYE2545
## DYE2545:AAACCCAAGGAAAGAC   DYE2545                <NA>             DYE2545
## DYE2545:AAACCCAAGGCTCTCG   DYE2545                <NA>             DYE2545
## DYE2545:AAACCCACAAACCACT   DYE2545                <NA>             DYE2545
## DYE2545:AAACCCACAAGTGGTG   DYE2545                <NA>             DYE2545
## DYE2545:AAACCCACAATACAGA   DYE2545                <NA>             DYE2545
##                          SRM_updated.1  SRM.1 SRM_Sample_ID_updated.1
## DYE2545:AAACCCAAGATGTTCC        841877 841877                 3251768
## DYE2545:AAACCCAAGGAAAGAC        841877 841877                 3251768
## DYE2545:AAACCCAAGGCTCTCG        841877 841877                 3251768
## DYE2545:AAACCCACAAACCACT        841877 841877                 3251768
## DYE2545:AAACCCACAAGTGGTG        841877 841877                 3251768
## DYE2545:AAACCCACAATACAGA        841877 841877                 3251768
##                          SRM_Sample_ID.1 Experimental_Sample.1 sample_3D.1
## DYE2545:AAACCCAAGATGTTCC         3251768  SJ1041CCLF 01.24.003          No
## DYE2545:AAACCCAAGGAAAGAC         3251768  SJ1041CCLF 01.24.003          No
## DYE2545:AAACCCAAGGCTCTCG         3251768  SJ1041CCLF 01.24.003          No
## DYE2545:AAACCCACAAACCACT         3251768  SJ1041CCLF 01.24.003          No
## DYE2545:AAACCCACAAGTGGTG         3251768  SJ1041CCLF 01.24.003          No
## DYE2545:AAACCCACAATACAGA         3251768  SJ1041CCLF 01.24.003          No
##                          sample_2D.NOT.NEEDED.2025.04.10.1 Tissue_Bank_ID.1
## DYE2545:AAACCCAAGATGTTCC                                No       TB-13-3627
## DYE2545:AAACCCAAGGAAAGAC                                No       TB-13-3627
## DYE2545:AAACCCAAGGCTCTCG                                No       TB-13-3627
## DYE2545:AAACCCACAAACCACT                                No       TB-13-3627
## DYE2545:AAACCCACAAGTGGTG                                No       TB-13-3627
## DYE2545:AAACCCACAATACAGA                                No       TB-13-3627
##                          Xenograft_CompBio_ID.1 CSTN_MAST_ID.1
## DYE2545:AAACCCAAGATGTTCC         SJNBL012407_X1         MAST97
## DYE2545:AAACCCAAGGAAAGAC         SJNBL012407_X1         MAST97
## DYE2545:AAACCCAAGGCTCTCG         SJNBL012407_X1         MAST97
## DYE2545:AAACCCACAAACCACT         SJNBL012407_X1         MAST97
## DYE2545:AAACCCACAAGTGGTG         SJNBL012407_X1         MAST97
## DYE2545:AAACCCACAATACAGA         SJNBL012407_X1         MAST97
##                                        Date.1 SRM_Submission_Information.1
## DYE2545:AAACCCAAGATGTTCC 2024-12-16T00:00:00Z         SJ1041CCLF 01.24.003
## DYE2545:AAACCCAAGGAAAGAC 2024-12-16T00:00:00Z         SJ1041CCLF 01.24.003
## DYE2545:AAACCCAAGGCTCTCG 2024-12-16T00:00:00Z         SJ1041CCLF 01.24.003
## DYE2545:AAACCCACAAACCACT 2024-12-16T00:00:00Z         SJ1041CCLF 01.24.003
## DYE2545:AAACCCACAAGTGGTG 2024-12-16T00:00:00Z         SJ1041CCLF 01.24.003
## DYE2545:AAACCCACAATACAGA 2024-12-16T00:00:00Z         SJ1041CCLF 01.24.003
##                          Compbioname_for_publications.1
## DYE2545:AAACCCAAGATGTTCC                 SJNBL012407_C3
```

```
## DYE2545:AAACCCAAGGAAAGAC                SJNBL012407_C3
## DYE2545:AAACCCAAGGCTCTCG                SJNBL012407_C3
## DYE2545:AAACCCACAAACCACT                SJNBL012407_C3
## DYE2545:AAACCCACAAGTGGTG                SJNBL012407_C3
## DYE2545:AAACCCACAATACAGA                SJNBL012407_C3
##                           Comp_Bio_Sample_Name.1 Comp_Bio_Subject_ID.1
## DYE2545:AAACCCAAGATGTTCC SJNBL012407_C3-SJ1041T_2             SJ012407
## DYE2545:AAACCCAAGGAAAGAC SJNBL012407_C3-SJ1041T_2             SJ012407
## DYE2545:AAACCCAAGGCTCTCG SJNBL012407_C3-SJ1041T_2             SJ012407
## DYE2545:AAACCCACAAACCACT SJNBL012407_C3-SJ1041T_2             SJ012407
## DYE2545:AAACCCACAAGTGGTG SJNBL012407_C3-SJ1041T_2             SJ012407
## DYE2545:AAACCCACAATACAGA SJNBL012407_C3-SJ1041T_2             SJ012407
##                          Preferred_Sample_ID.1 Sample_Type_Code.1
## DYE2545:AAACCCAAGATGTTCC             SJ1041T_2                  C
## DYE2545:AAACCCAAGGAAAGAC             SJ1041T_2                  C
## DYE2545:AAACCCAAGGCTCTCG             SJ1041T_2                  C
## DYE2545:AAACCCACAAACCACT             SJ1041T_2                  C
## DYE2545:AAACCCACAAGTGGTG             SJ1041T_2                  C
## DYE2545:AAACCCACAATACAGA             SJ1041T_2                  C
##                          Prior_Comp_Bio_ID.1 PATH.1 NOTES.1 match_SJID_RPMI.1
## DYE2545:AAACCCAAGATGTTCC         SJNBL012407   <NA>    <NA>    SJNBL012407_no
## DYE2545:AAACCCAAGGAAAGAC         SJNBL012407   <NA>    <NA>    SJNBL012407_no
## DYE2545:AAACCCAAGGCTCTCG         SJNBL012407   <NA>    <NA>    SJNBL012407_no
## DYE2545:AAACCCACAAACCACT         SJNBL012407   <NA>    <NA>    SJNBL012407_no
## DYE2545:AAACCCACAAGTGGTG         SJNBL012407   <NA>    <NA>    SJNBL012407_no
## DYE2545:AAACCCACAATACAGA         SJNBL012407   <NA>    <NA>    SJNBL012407_no
##                           S.Score.1 G2M.Score.1 Phase.1  UMAP_1.x   UMAP_2.x
## DYE2545:AAACCCAAGATGTTCC -0.22486652  -0.2675636      G1  9.085792 -1.5443104
## DYE2545:AAACCCAAGGAAAGAC -0.05736943  -0.1675573      G1  6.835735 -2.2798051
## DYE2545:AAACCCAAGGCTCTCG -0.20725037  -0.1973390      G1  5.306400 -2.2335602
## DYE2545:AAACCCACAAACCACT -0.28974713  -0.2804960      G1  4.128226 -0.2242093
## DYE2545:AAACCCACAAGTGGTG -0.31975873  -0.3241192      G1  4.214514 -1.3846704
## DYE2545:AAACCCACAATACAGA -0.11510892  -0.2054158      G1 -3.577932  2.3944530
##                             PC_1.x      PC_2.x    PC_3.x     PC_4.x    PC_5.x
## DYE2545:AAACCCAAGATGTTCC -15.863874 -32.6546839 0.8260766 -18.296167 7.4672071
## DYE2545:AAACCCAAGGAAAGAC  -9.431601 -21.4253276 1.4509622 -15.964464 2.2837768
## DYE2545:AAACCCAAGGCTCTCG  -9.189331 -11.7659123 2.8201872 -12.901577 3.0042420
## DYE2545:AAACCCACAAACCACT  -0.632111  -8.1764752 4.6344284 -11.745347 1.4077577
## DYE2545:AAACCCACAAGTGGTG  -3.746334  -7.1768090 2.1580190 -12.186304 0.3765286
## DYE2545:AAACCCACAATACAGA  -4.619726   0.4027877 3.6466166  -5.877808 3.7909257
##                            PC_6.x     PC_7.x    PC_8.x     PC_9.x     PC_10.x
## DYE2545:AAACCCAAGATGTTCC -6.495609  6.9242178 10.356101 -3.1697889 -22.8791552
## DYE2545:AAACCCAAGGAAAGAC -5.869902  3.2564196  3.783053 -4.3649521   0.4776367
## DYE2545:AAACCCAAGGCTCTCG  1.099606 -1.0354393 -4.115567 -2.4106450   6.4005139
## DYE2545:AAACCCACAAACCACT  1.638154 -0.9564433 -7.009008 -3.0709138  13.9329421
## DYE2545:AAACCCACAAGTGGTG  1.134238 -0.7682338 -7.173833 -2.4168836  10.5425499
## DYE2545:AAACCCACAATACAGA  5.151632 -2.8199820 -5.585466  0.1286505   5.3019733
##                           PC_11.x   PC_12.x    PC_13.x    PC_14.x    PC_15.x
## DYE2545:AAACCCAAGATGTTCC  2.318805  9.863862 -7.4874913 10.7100504 -6.3503098
## DYE2545:AAACCCAAGGAAAGAC  1.408204 -2.874774  3.5337765 -2.1880440  1.2392725
## DYE2545:AAACCCAAGGCTCTCG -2.511625 -4.417163  3.4626152 -5.3307465  3.8559767
```

```
## DYE2545:AAACCCACAAACCACT -3.885880  2.272538 -0.6860176  3.0788362 -1.7560677
## DYE2545:AAACCCACAAGTGGTG -1.590351 -2.827610 -1.4508638  2.0134072 -0.3433200
## DYE2545:AAACCCACAATACAGA -3.024046  2.045094 -1.1220299  0.4380948 -0.5391842
##                             PC_16.x    PC_17.x     PC_18.x      PC_19.x
## DYE2545:AAACCCAAGATGTTCC  2.2426751  1.8916413  0.75978639 -0.788243693
## DYE2545:AAACCCAAGGAAAGAC -1.2604325 -0.9685792 -3.78744852  0.692745572
## DYE2545:AAACCCAAGGCTCTCG -1.1604766 -1.7994313 -2.49335533 -1.601766611
## DYE2545:AAACCCACAAACCACT  0.6154134  1.6720582  0.07299006  1.929624380
## DYE2545:AAACCCACAAGTGGTG -1.1592913  2.1567640 -4.65308816 -2.328754025
## DYE2545:AAACCCACAATACAGA  0.2028160 -0.4331900  0.95269650  0.005967839
##                             PC_20.x     PC_21.x    PC_22.x    PC_23.x
## DYE2545:AAACCCAAGATGTTCC -2.2202915 -2.38143583 -2.9730362 -0.9262151
## DYE2545:AAACCCAAGGAAAGAC  3.5548754  2.98895923  2.4440021  0.9979305
## DYE2545:AAACCCAAGGCTCTCG -0.9419118  2.85820635  0.3180130 -0.5923591
## DYE2545:AAACCCACAAACCACT  0.7687017 -0.08372203  0.4231828 -0.7862789
## DYE2545:AAACCCACAAGTGGTG -1.1156528 -1.02724389  2.5277115 -1.7370136
## DYE2545:AAACCCACAATACAGA -1.6297936 -0.53810839 -1.0408624 -2.2783067
##                             PC_24.x     PC_25.x    PC_26.x    PC_27.x
## DYE2545:AAACCCAAGATGTTCC  3.5626954  6.52893164 -2.1101894  0.5164932
## DYE2545:AAACCCAAGGAAAGAC -0.5610950  0.04656445  1.5751073 -2.6024575
## DYE2545:AAACCCAAGGCTCTCG -0.9212251 -3.33533561  0.4304649 -0.9915104
## DYE2545:AAACCCACAAACCACT -1.2788869 -1.49469591  0.2314583  0.8904170
## DYE2545:AAACCCACAAGTGGTG -0.2650034 -2.29214870 -0.4044628 -0.2296887
## DYE2545:AAACCCACAATACAGA  0.5782148 -1.53033807  0.1998572  0.8766829
##                              PC_28.x    PC_29.x    PC_30.x RNA_snn_res.0.10
## DYE2545:AAACCCAAGATGTTCC  2.65092355 -6.4706949 -2.1001067                1
## DYE2545:AAACCCAAGGAAAGAC  0.36146655  0.6870124  0.8580130                1
## DYE2545:AAACCCAAGGCTCTCG  0.66481255 -3.3209920  4.7293540                1
## DYE2545:AAACCCACAAACCACT  0.01733426  1.4488693 -2.5102127                1
## DYE2545:AAACCCACAAGTGGTG -1.35753530  0.8580106  2.0390308                1
## DYE2545:AAACCCACAATACAGA -1.53763292  1.8993305  0.0598051                1
##                          seurat_clusters.1 RNA_snn_res.0.1.1 RNA_snn_res.0.2.1
## DYE2545:AAACCCAAGATGTTCC                15                 3                 4
## DYE2545:AAACCCAAGGAAAGAC                 7                 3                 4
## DYE2545:AAACCCAAGGCTCTCG                54                 3                 4
## DYE2545:AAACCCACAAACCACT                44                 3                 4
## DYE2545:AAACCCACAAGTGGTG                44                 3                 4
## DYE2545:AAACCCACAATACAGA                17                 2                 2
##                          RNA_snn_res.0.3.1 RNA_snn_res.0.4.1 RNA_snn_res.0.5.1
## DYE2545:AAACCCAAGATGTTCC                 5                 6                 6
## DYE2545:AAACCCAAGGAAAGAC                 5                 6                 6
## DYE2545:AAACCCAAGGCTCTCG                 5                 7                 7
## DYE2545:AAACCCACAAACCACT                 5                 7                 7
## DYE2545:AAACCCACAAGTGGTG                 5                 7                 7
## DYE2545:AAACCCACAATACAGA                 1                 2                 1
##                          RNA_snn_res.0.6.1 RNA_snn_res.0.7.1 RNA_snn_res.0.8.1
## DYE2545:AAACCCAAGATGTTCC                 8                 9                 9
## DYE2545:AAACCCAAGGAAAGAC                 8                 9                 9
## DYE2545:AAACCCAAGGCTCTCG                 5                 7                 7
## DYE2545:AAACCCACAAACCACT                 5                 7                 7
## DYE2545:AAACCCACAAGTGGTG                 5                 7                 7
```

```
## DYE2545:AAACCCACAATACAGA                 1                 2                 2
##                           RNA_snn_res.0.9.1 RNA_snn_res.1.1 RNA_snn_res.2.1
## DYE2545:AAACCCAAGATGTTCC                 8               8              23
## DYE2545:AAACCCAAGGAAAGAC                 8               8              11
## DYE2545:AAACCCAAGGCTCTCG                 6               6               1
## DYE2545:AAACCCACAAACCACT                 6               6               1
## DYE2545:AAACCCACAAGTGGTG                 6               6               1
## DYE2545:AAACCCACAATACAGA                 2               2               2
##                           RNA_snn_res.3.1 RNA_snn_res.4.1 RNA_snn_res.5.1
## DYE2545:AAACCCAAGATGTTCC              26              23              19
## DYE2545:AAACCCAAGGAAAGAC               6               6              10
## DYE2545:AAACCCAAGGCTCTCG              20              22              32
## DYE2545:AAACCCACAAACCACT               7              20              14
## DYE2545:AAACCCACAAGTGGTG               7              20              14
## DYE2545:AAACCCACAATACAGA               1               4               2
##                           RNA_snn_res.6.1 RNA_snn_res.7.1 RNA_snn_res.8.1
## DYE2545:AAACCCAAGATGTTCC               8               4              19
## DYE2545:AAACCCAAGGAAAGAC               5              15               6
## DYE2545:AAACCCAAGGCTCTCG              29              25              20
## DYE2545:AAACCCACAAACCACT              13              16              52
## DYE2545:AAACCCACAAGTGGTG              13              16              52
## DYE2545:AAACCCACAATACAGA               1               1              32
##                           RNA_snn_res.9.1 RNA_snn_res.10.1  UMAP_1.y  UMAP_2.y
## DYE2545:AAACCCAAGATGTTCC              15               15 -4.479104 -
9.875414
## DYE2545:AAACCCAAGGAAAGAC              11                7 -1.760218 -
9.614166
## DYE2545:AAACCCAAGGCTCTCG              19               54 -1.269360 -
7.994416
## DYE2545:AAACCCACAAACCACT              48               44  2.468587 -
7.785643
## DYE2545:AAACCCACAAGTGGTG              48               44  1.326973 -
8.561863
## DYE2545:AAACCCACAATACAGA               1               17  0.594964 -
3.794753
##                              PC_1.y     PC_2.y     PC_3.y     PC_4.y     PC_5.y
## DYE2545:AAACCCAAGATGTTCC -44.361736 -0.3498955 -6.5871771  21.313161  12.820344
## DYE2545:AAACCCAAGGAAAGAC -29.943065  0.1864216 -7.1310710   2.413288   8.934736
## DYE2545:AAACCCAAGGCTCTCG -21.703032 -2.2783137  0.4004396  -2.087764  -
5.307791
## DYE2545:AAACCCACAAACCACT -16.056743 -3.7813880  1.6416767 -17.904250  -
5.240698
## DYE2545:AAACCCACAAGTGGTG -15.417941 -0.1092811  0.7468595 -13.424479  -
4.851114
## DYE2545:AAACCCACAATACAGA  -7.529097 -5.4293654  6.0714155  -7.361042 -
10.553899
##                              PC_6.y     PC_7.y     PC_8.y     PC_9.y    PC_10.y
## DYE2545:AAACCCAAGATGTTCC  15.726013  4.2116332 -21.558956  4.9322034 -
0.8223053
## DYE2545:AAACCCAAGGAAAGAC  -3.828320 -2.1594133   4.781474 -4.7728300  5.3965381
## DYE2545:AAACCCAAGGCTCTCG  -6.335432 -1.2832691   9.624468 -0.1125284  3.2770683
```

```
## DYE2545:AAACCCACAAACCACT -13.277945 -0.2213529  -4.942191 -1.3347567 -
1.1885515
## DYE2545:AAACCCACAAGTGGTG  -8.869318 -1.8440749   1.294917  4.1570867  4.6242295
## DYE2545:AAACCCACAATACAGA  -4.874613  3.2249348  -1.943663  0.6265535 -
1.6139765
##                             PC_11.y    PC_12.y    PC_13.y    PC_14.y   PC_15.y
## DYE2545:AAACCCAAGATGTTCC  5.440777  5.8972868 -5.5928338 10.3379803  3.248851
## DYE2545:AAACCCAAGGAAAGAC -1.056980 -0.5824754 -0.8486377 -1.3789469  1.703671
## DYE2545:AAACCCAAGGCTCTCG -1.139655 -3.7065993 -1.8451374 -0.4306136 -
2.347052
## DYE2545:AAACCCACAAACCACT  1.235513 -1.0370845  1.9786905 -2.9240144 -
2.333288
## DYE2545:AAACCCACAAGTGGTG  3.758541 -4.2129357  1.9023722  0.7556897 -
5.665886
## DYE2545:AAACCCACAATACAGA  1.163025 -0.2710679  1.8318496 -2.6683271 -
1.655289
##                             PC_16.y    PC_17.y    PC_18.y     PC_19.y    PC_20.y
## DYE2545:AAACCCAAGATGTTCC -4.181243 -0.2007954  7.0198663 -2.13214100 -
2.0616200
## DYE2545:AAACCCAAGGAAAGAC  3.742322  1.1910307 -5.5069176 -2.91120879  2.0355320
## DYE2545:AAACCCAAGGCTCTCG  2.964191 -7.6537637  2.5299749  1.48133043  0.9937034
## DYE2545:AAACCCACAAACCACT -0.899313  2.1578213  0.8875158 -0.98371927  0.3034186
## DYE2545:AAACCCACAAGTGGTG -1.734138 -1.9177623  2.4455817  0.05078623 -
1.7541679
## DYE2545:AAACCCACAATACAGA -2.849648  1.0050598 -0.8500319 -0.30100498 -
0.9873374
##                              PC_21.y    PC_22.y   PC_23.y     PC_24.y    PC_25.y
## DYE2545:AAACCCAAGATGTTCC -1.6662457  5.1392864 -1.357076  2.29986514  1.0475817
## DYE2545:AAACCCAAGGAAAGAC  1.4015604 -3.3642872  1.586199  3.11939815 -
2.7687263
## DYE2545:AAACCCAAGGCTCTCG -2.2680682  0.8736335  1.080483  1.43176329 -
1.4161921
## DYE2545:AAACCCACAAACCACT -2.4006588  2.3422282 -3.614108  3.87268282 -
3.8345919
## DYE2545:AAACCCACAAGTGGTG  0.7149712  6.0929420 -5.676803 -0.03936357  0.4923386
## DYE2545:AAACCCACAATACAGA -1.8875031  1.8841155 -4.504857  1.70250816 -
1.0875413
##                               PC_26.y     PC_27.y    PC_28.y   PC_29.y   PC_30.y
## DYE2545:AAACCCAAGATGTTCC  0.44818457 -1.49122713 -0.2263842 -3.329649  2.933700
## DYE2545:AAACCCAAGGAAAGAC -0.03660816  0.32367936 -0.5247622  2.407370 -
1.988673
## DYE2545:AAACCCAAGGCTCTCG  1.04318230  1.52810102  1.1003517  1.065225  1.645771
## DYE2545:AAACCCACAAACCACT  0.48679263 -0.08399117  0.6750942 -1.574071  6.020454
## DYE2545:AAACCCACAAGTGGTG -3.52152204  0.24052152 -2.7119793 -1.015690 -
1.281054
## DYE2545:AAACCCACAATACAGA  1.48660300 -1.32376725  1.3599089 -0.970910  1.268743
```

```r
saveRDS(seurat_obj, file = paste0(results_dir, "/", glue::glue("seurat_obj_integrated_{inte
```

10 Session Info

```
## R version 4.4.0 (2024-04-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK versi
##
## locale:
## [1] C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
## [1] grid      stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] knitr_1.49           reshape2_1.4.4      SeuratWrappers_0.2.0
##  [4] RcppPlanc_1.0.0      rliger_2.1.0.9004   harmony_1.2.3
##  [7] Rcpp_1.0.14          future_1.34.0       RColorBrewer_1.1-3
## [10] ggthemes_5.1.0       patchwork_1.3.0     scooter_0.0.0.9004
## [13] SeuratObject_5.0.2   Seurat_4.4.0        lubridate_1.9.4
## [16] forcats_1.0.0        stringr_1.5.1       dplyr_1.1.4
## [19] purrr_1.0.2          readr_2.1.5         tidyr_1.3.1
## [22] tibble_3.2.1         ggplot2_3.5.1       tidyverse_2.0.0
## [25] yaml_2.3.10
##
## loaded via a namespace (and not attached):
##   [1] jsonlite_1.8.9         magrittr_2.0.3         ggbeeswarm_0.7.2
##   [4] spatstat.utils_3.1-2   farver_2.1.2           rmarkdown_2.29
##   [7] ragg_1.3.3             vctrs_0.6.5            ROCR_1.0-11
##  [10] spatstat.explore_3.3-4 tinytex_0.54           htmltools_0.5.8.1
##  [13] sass_0.4.9             sctransform_0.4.1      parallelly_1.42.0
##  [16] bslib_0.9.0            KernSmooth_2.23-26     htmlwidgets_1.6.4
##  [19] ica_1.0-3              plyr_1.8.9             cachem_1.1.0
##  [22] plotly_4.10.4          zoo_1.8-12             igraph_2.1.4
##  [25] mime_0.12              lifecycle_1.0.4        pkgconfig_2.0.3
##  [28] rsvd_1.0.5             Matrix_1.7-2           R6_2.5.1
##  [31] fastmap_1.2.0          fitdistrplus_1.2-2     shiny_1.10.0
##  [34] digest_0.6.37          colorspace_2.1-1       S4Vectors_0.44.0
##  [37] tensor_1.5             irlba_2.3.5.1          textshaping_1.0.0
##  [40] labeling_0.4.3         progressr_0.15.1       spatstat.sparse_3.1-0
##  [43] timechange_0.3.0       httr_1.4.7             polyclip_1.10-7
##  [46] abind_1.4-8            compiler_4.4.0         remotes_2.5.0
##  [49] bit64_4.6.0-1          withr_3.0.2            MASS_7.3-64
##  [52] ggsci_3.2.0            tools_4.4.0            vipor_0.4.7
##  [55] lmtest_0.9-40          beeswarm_0.4.0         httpuv_1.6.15
```

```
##  [58] future.apply_1.11.3    goftest_1.2-3          glue_1.8.0
##  [61] nlme_3.1-167           promises_1.3.2         Rtsne_0.17
##  [64] cluster_2.1.8          generics_0.1.3         gtable_0.3.6
##  [67] spatstat.data_3.1-4    tzdb_0.4.0             data.table_1.16.4
##  [70] hms_1.1.3              sp_2.2-0               BiocGenerics_0.52.0
##  [73] spatstat.geom_3.3-5    RcppAnnoy_0.0.22       ggrepel_0.9.6
##  [76] RANN_2.6.2             pillar_1.10.1          spam_2.11-1
##  [79] vroom_1.6.5            later_1.4.1            splines_4.4.0
##  [82] lattice_0.22-6         survival_3.8-3         bit_4.5.0.1
##  [85] deldir_2.0-4           tidyselect_1.2.1       miniUI_0.1.1.1
##  [88] pbapply_1.7-2          gridExtra_2.3          scattermore_1.2
##  [91] RhpcBLASctl_0.23-42    stats4_4.4.0           xfun_0.50
##  [94] matrixStats_1.5.0      stringi_1.8.4          lazyeval_0.2.2
##  [97] evaluate_1.0.3         codetools_0.2-20       BiocManager_1.30.25
## [100] cli_3.6.3              uwot_0.2.2             xtable_1.8-4
## [103] reticulate_1.40.0      systemfonts_1.2.1      jquerylib_0.1.4
## [106] munsell_0.5.1          globals_0.16.3         spatstat.random_3.3-2
## [109] png_0.1-8              ggrastr_1.0.2          spatstat.univar_3.1-1
## [112] parallel_4.4.0         dotCall64_1.2          listenv_0.9.1
## [115] viridisLite_0.4.2      scales_1.3.0           ggridges_0.5.6
## [118] leiden_0.4.3.1         crayon_1.5.3           rlang_1.1.5
## [121] cowplot_1.1.3
```